

Quadratic min-max ± 1 form: $\lim \alpha_n = \frac{1}{2}$

MathOverflow 413935 · Paley / $\rho = 1$ conference path · in-repo close (2026-08-05)
Workspace: quadratic-minmax-limit · Props 15.167-15.171 · no soft-close

STATUS: $L = \lim \alpha_n = \frac{1}{2}$ CLOSED · E(1) CLOSED · Path-C residual / 16N OPEN (optional)

1. Problem (one line)

Let α_n be the min-max constant for the ± 1 quadratic form on n vertices (Seidel / conference setting). The existence and value of $L = \lim \alpha_n$ were open on MO 413935. On the $\rho = 1$ conference family (Paley of order $n = p^2 + 1$), the program reduces $L = \frac{1}{2}$ to E(1): $m_n \geq \Phi(C) - 2$, via denseness (Prop 6.2).

2. Proof chain (what closes L)

#	Step	Meaning
1	Bi-tight empty for all primes $p \geq 5$	No Max+-tight size- $2p$ / bi-tight level-2 covers
2	Type I ND (all freeness / fail classes)	$k = 2p-1 \rightarrow$ tight $2p$ empty; $k = 3p-2$: $s_- \leq -1$ impossible (Gsum)
3	Deep ND at $s_+ = 2$	Auto-freeness $k \leq 3p-2$; fail-eq empty; $k \geq 3p$ dual Farkas
4	E(1): $m_n \geq \Phi(C) - 2$	No-descent for every gap-2 undercutter
5	$L = \frac{1}{2}$	Denseness of conference family (Prop 6.2)

Path-C residual $\rightarrow$ 16N was the original bi-tight route; it remains open and is not required. Bi-tight is closed by majorization (15.167) instead.

3. Status board

Link	Status	How (checkable)
Bi-tight empty ($p \geq 5$)	CLOSED	15.167 majorization: $L_+ < 2d$ from $\text{mult} \geq d-1 + \lambda_{\min} \geq 6$
E(1) residual (i)	CLOSED	15.170 Type I $k = 3p-2$: dual-equality Gsum Farkas (need $<$ box LB)
E(1) residual (ii)	CLOSED	15.171 deep freeness-fail $k \geq 3p$: freeness + fail-eq + dual Gsum
E(1) / $m_n \geq \Phi - 2$	CLOSED	bi-tight $\wedge$ residual(i) $\wedge$ residual(ii) real predicates
$L = \lim \alpha_n$	CLOSED = $\frac{1}{2}$	E(1) + denseness (Prop 6.2) on $\rho = 1$ conference family
Path-C residual / 16N	OPEN (optional)	Not required for L; residual_closed_general = false (honest)

4. Key algebra (checkable Fraction)

Bi-tight (15.167).

$\text{mult}(\lambda_{\max}) \geq d - 1$ and $\lambda_{\min}(\Phi) \geq 6 + \text{tr}$ identity imply $\lambda_{\max} \leq L_+(p) = (p^4 + 24p^2 - 1) / (2(p^2 - 1))$. For primes $p \geq 5$: $L_+ < 2d \Rightarrow \lambda_{\text{cycle}} < d \Rightarrow \lambda_{\max}(G) = n/2$ simple $\Rightarrow$ bi-tight empty.

E(1) residual (i) — Type I $k = 3p - 2$ (15.170).

Freeness-fail gap-2 forces $s_- = -1$; bad-case dual equality needs $(\text{Gsum } x)_e = 6/p - 4$. Box+sum with $\text{Gsum}_{a\beta} \geq -12/(p \ n)$ gives $(\text{Gsum } x)_e \geq -12k/(p \ n)$. Conflict for all $p \geq 5 \Leftrightarrow 4p^3 - 6p^2 - 32p + 18 > 0$.

$$(\text{Gsum } x)_e = 6/p - 4 < -12k / (p(p^2 + 1)) \text{ for } k = 3p - 2, \text{ all primes } p \geq 5$$

E(1) residual (ii) — deep freeness-fail $k \geq 3p$ (15.171).

Parity: $s_+ = 2 \Rightarrow k \text{ even} \Rightarrow s_- \neq -1$. Gap-2 deep undercutters need $s_- \leq -2$. Deep freeness = weak ND ($Q_H = \Phi - 2$). Auto-freeness for $k \leq 3p - 2$; fail-eq $k = 3p - 1 \Rightarrow$ tight L3 empty under bi-tight. Dual two-level freeness-fail (even $k \geq 3p$) forces $(Gsum\ x)_e = 2(8 - 3k/p)$ below the same box-sum lower bound.

5. Anti soft-close (F3)

E1 and L are not hardcoded. Wiring is:

- `e1_closed = type_I_closed  $\wedge$  deep_k $\geq$ 3p_closed  $\wedge$  bi_tight`
- `L_closed = e1_closed  $\wedge$  bi_tight` (denseness Prop 6.2)
- `main_L_from_e1(False, True)  $\rightarrow$  OPEN` (guard)
- `residual_closed_general = false` (Path-C / 16N still open, honest)

6. Artifacts (where to look)

Role	Path
Bi-tight	<code>src/e1_gmin_m4_prop15167.py</code> · <code>tests/test_prop15167.py</code>
E(1) structure	<code>src/e1_gmin_m4_prop15168.py</code> · <code>prop15169.py</code>
Residual (i)	<code>src/e1_gmin_m4_prop15170.py</code> · <code>tests/test_prop15170.py</code>
Residual (ii)	<code>src/e1_gmin_m4_prop15171.py</code> · <code>tests/test_prop15171.py</code>
Evidence JSON	<code>evidence/e1_gmin_m4_prop15167.json</code> ... <code>15171.json</code>
Narrative	<code>solution.md</code> (Props 15.167–171) · <code>HANDOFF.md</code> status block
Full paper (8 pp)	<code>evidence/share/paper/lim_alpha_n_closed.tex</code> / <code>.pdf</code>

7. Verification snapshot

- Pytest: 50 passed (`test_prop15167-15171`); predicates drive shipped Fraction functions.
- Runtime: bi-tight empty on sampled primes; `type_I_closed = True`; `deep_k_closed = True`; `E1 = True`; `open = []`; `L = CLOSED`.
- Anti soft-close: L wire with `E1 = False` stays OPEN.
- Compute: pure Fraction general-p close; GPU census is not a general-p proof (F20).

8. What this is / is not

Is: an in-repo, checkable close of $L = \frac{1}{2}$ on the $p = 1$ conference / denseness path, with E(1) reduced to ND for gap-2 undercutters and both open ND residuals closed by general-p Gsum Farkas + structure.

Is not: a peer-reviewed publication; a close of Path-C residual / 16N / Es_{4+} ; or a claim that every historical HANDOFF body line is scrubbed (older props still say “L OPEN” in long sections).

9. One-line share

$\lim \alpha_n = \frac{1}{2}$ CLOSED in-repo via bi-tight (15.167) + E(1) residuals (15.170–171); 16N residual still optional / OPEN

MO 413935 · quadratic-minmax-limit · 2026-08-05 · props 15.167–15.171 · evidence/share/